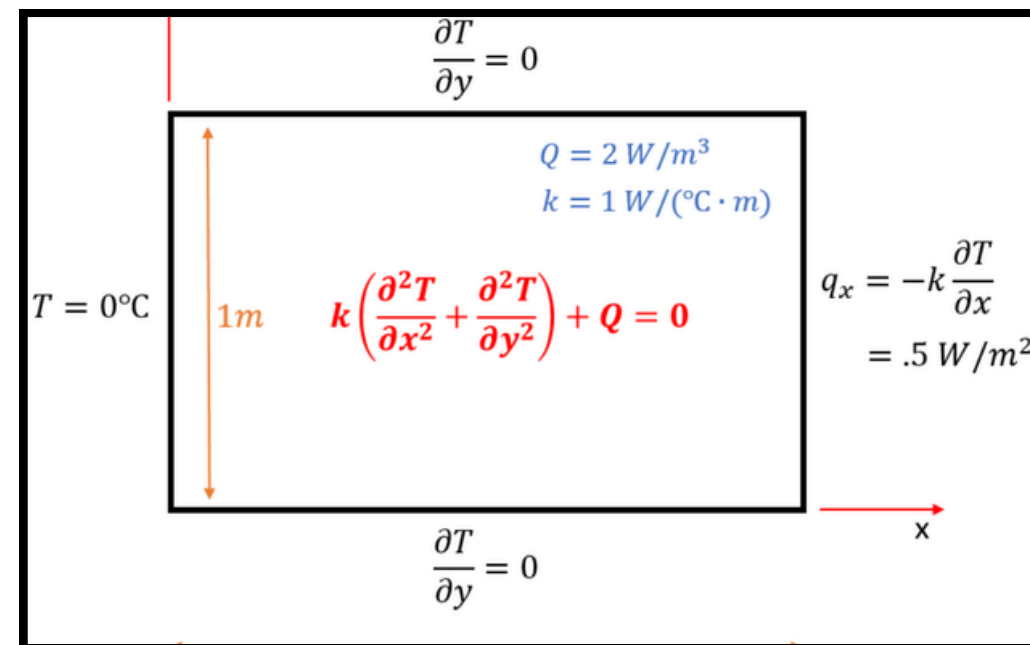
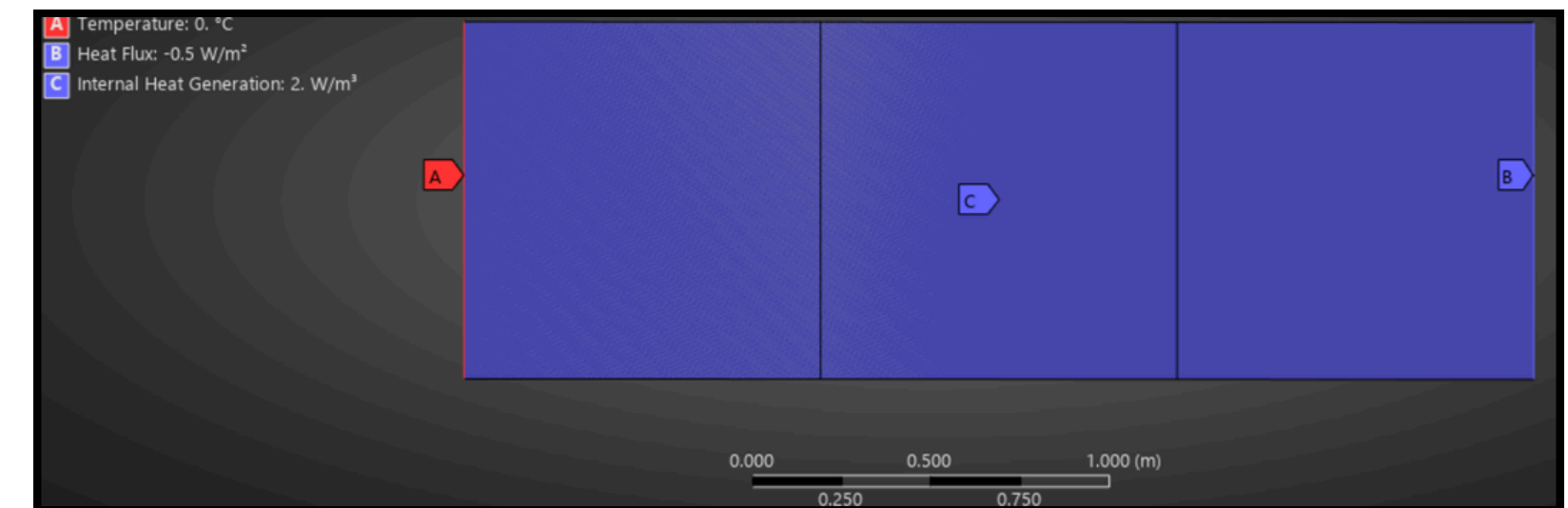


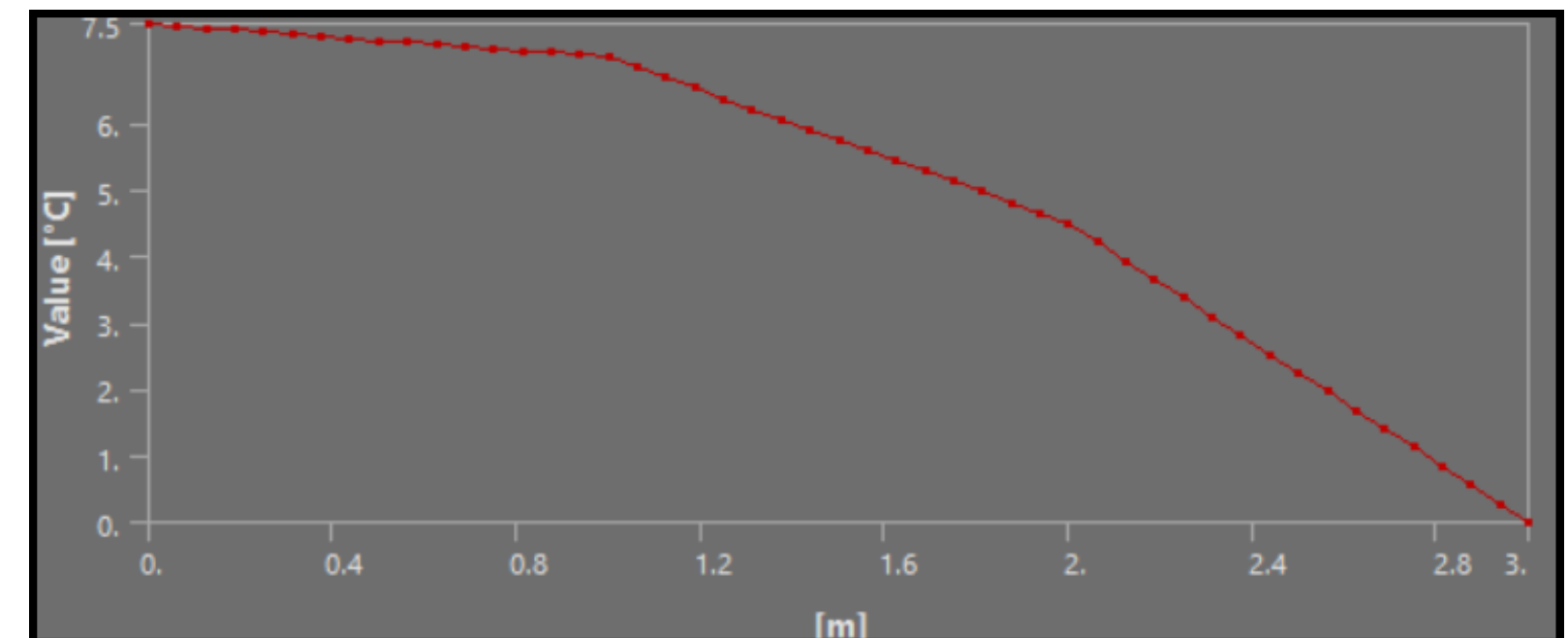
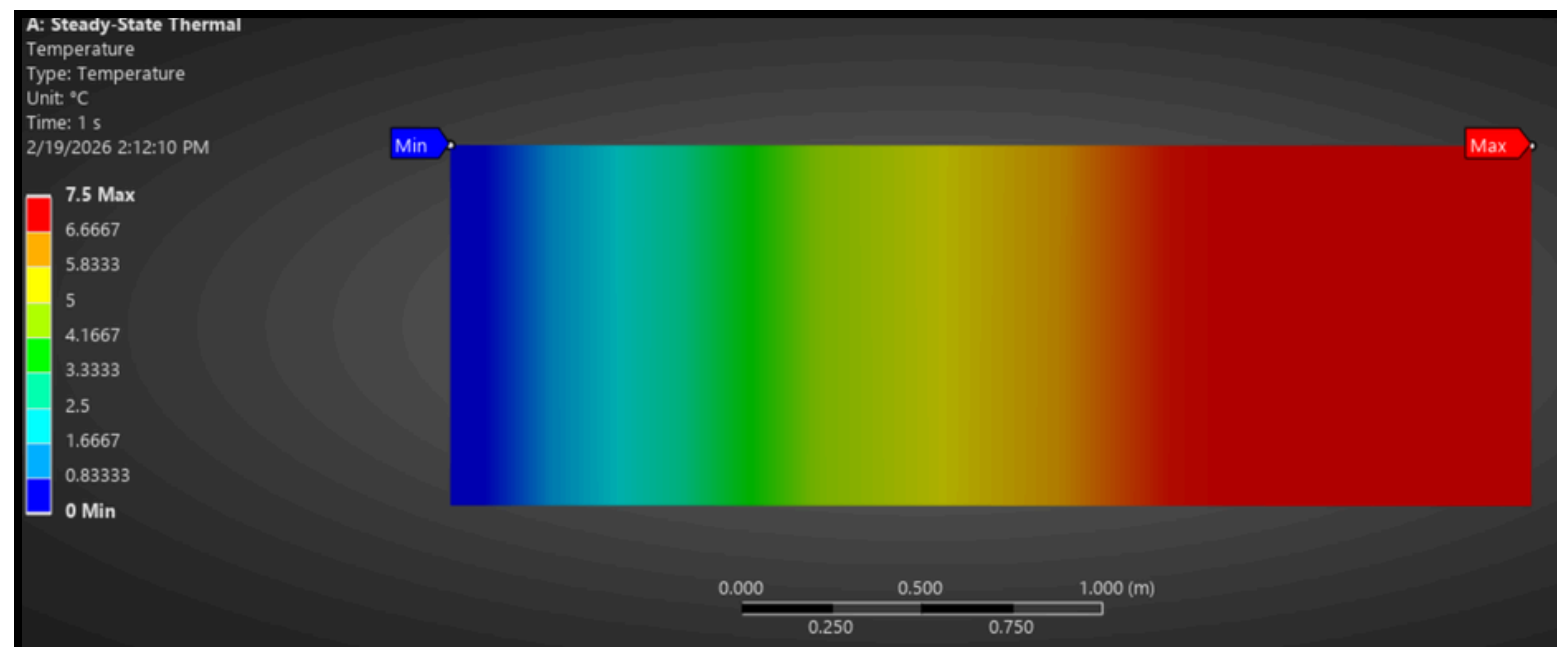
Steady State Thermal Analysis - 2D Conduction



Heat transfer reference for Ansys models. Inputs are insulated boundary conditions, constant temp, heat flux, and internal generation.

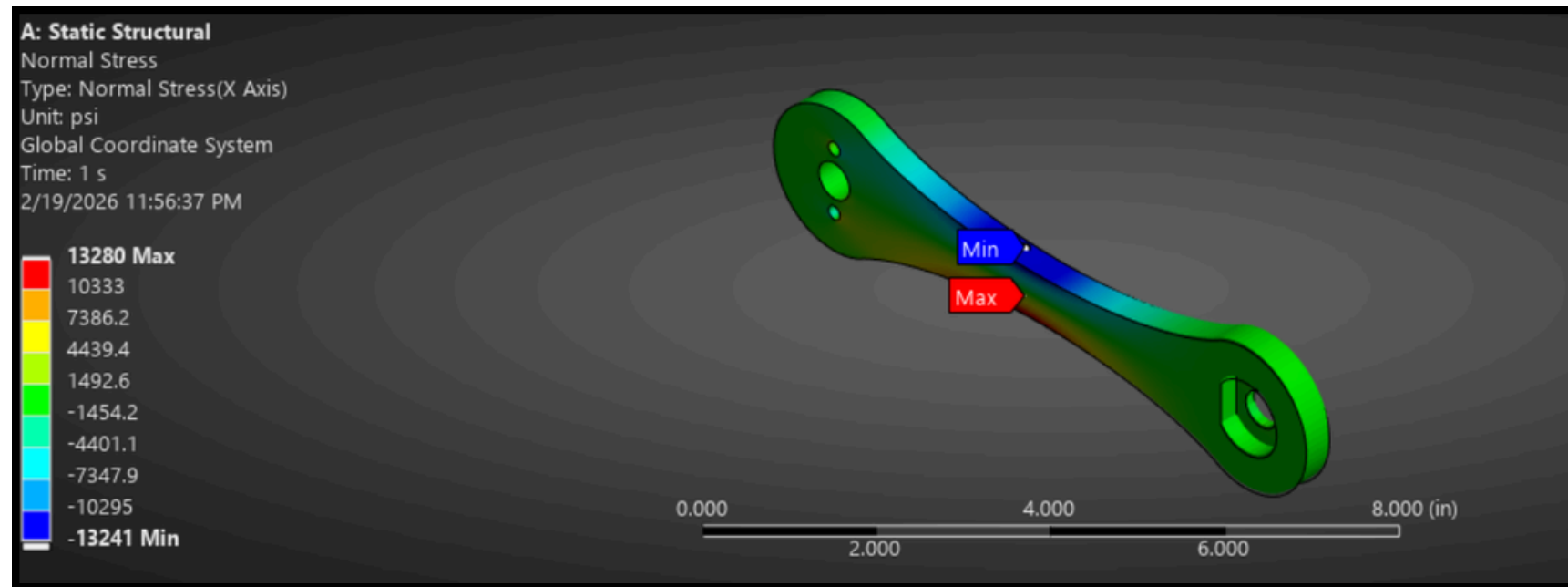


Model setup including thermal inputs, and simple mesh design to probe temperatures at nodes after generating data.



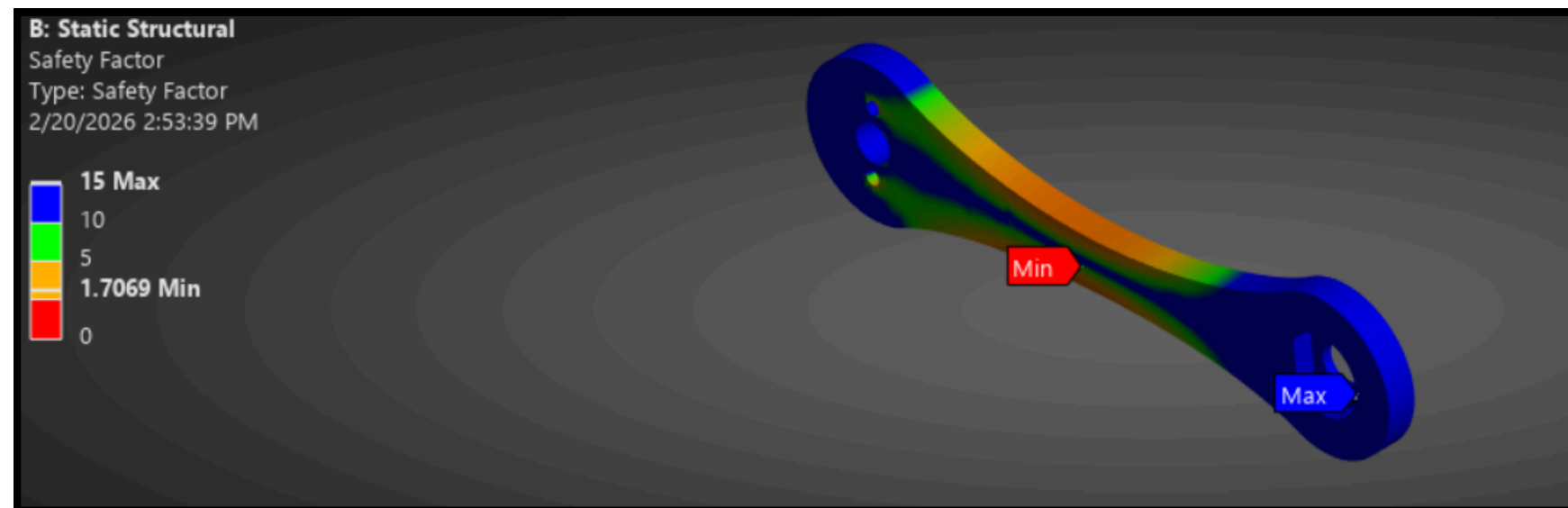
Temperature vs Distance based on path data along surface

Bike Crank Static Structural Analysis of Stress

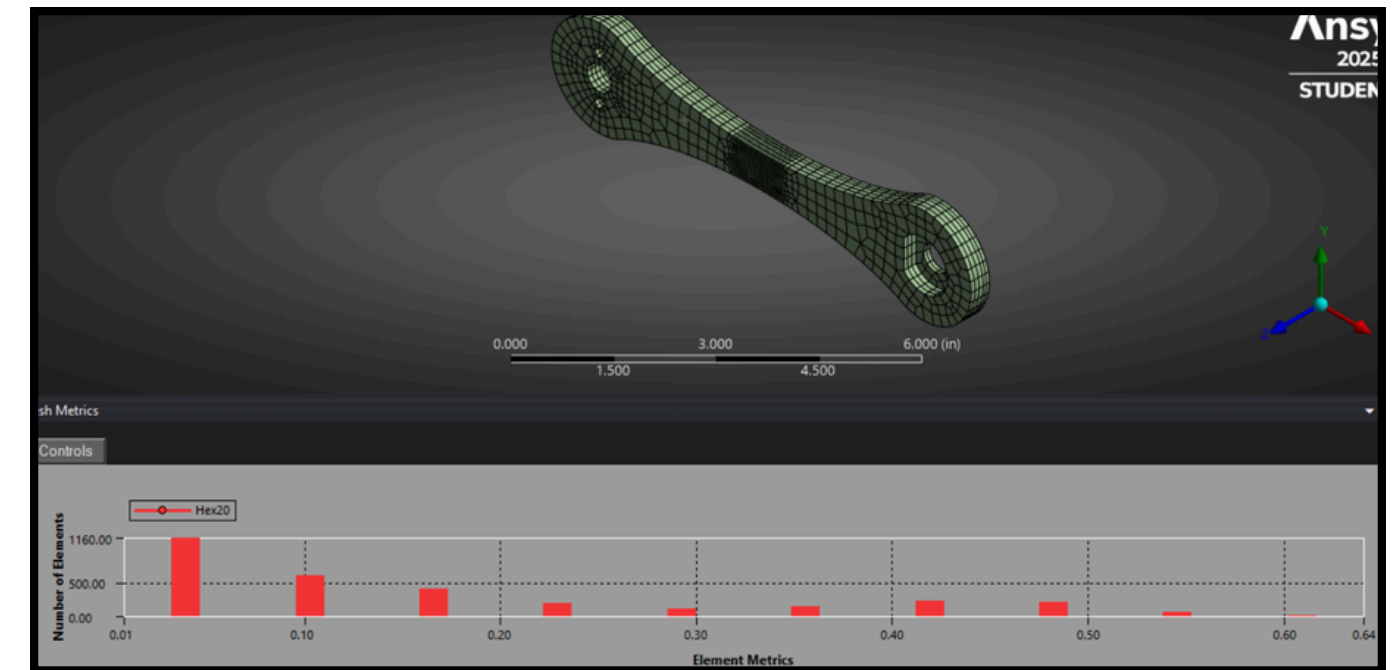


Al 6061 bike crank with 100 lbf applied load in y direction and fixed surface to test stresses and deformation. Verified by:

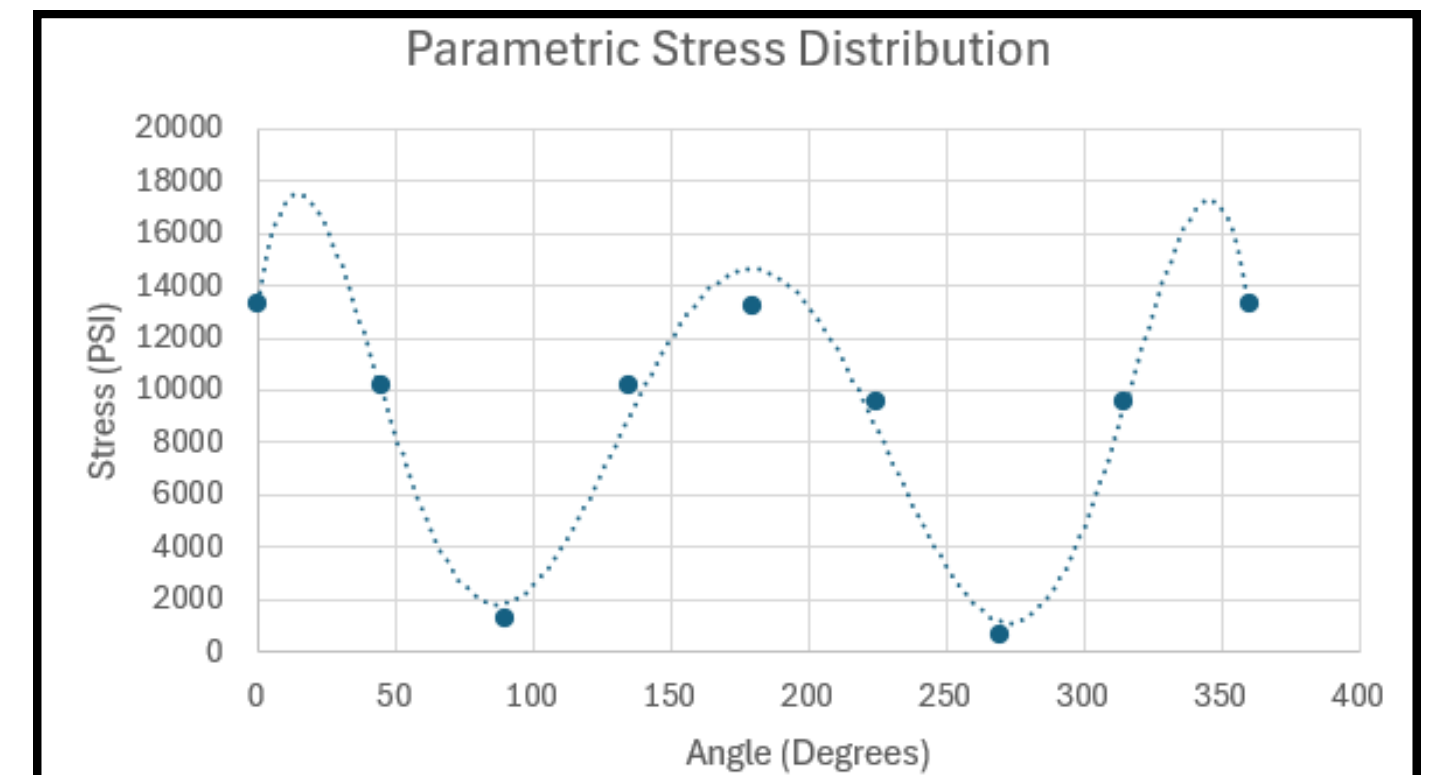
$$\sigma = -My/I = \frac{\frac{6.7}{2} (100) (0.325)}{\frac{0.375(0.65^3)}{12}} = 12,686 \text{ PSI } z=0.375, y = 0.65 \text{ } x = 4 \text{ at NA}$$



Simple FOS fatigue model for 1e7 cycle under zero based Goodman conditions. Validates expected outcome for a real engineered bike crank.



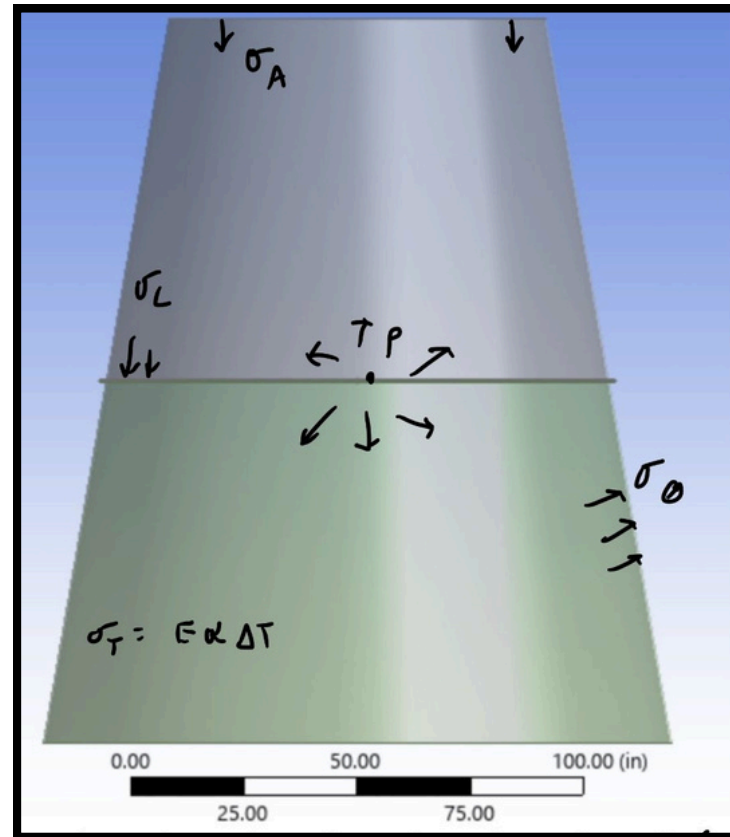
Local SOI mesh refinement to test mesh convergence and skewness to confirm mesh quality.



Parametric stress distribution modeling sine and cosine components of force if crank rotates.

Brycen Schoenberg

Bolted Flange Thermal, Pressure and Force Conditions



For stainless steel, $\sigma_r = \alpha E \Delta T = (9.5e^{-6} 1/F)(2.9e^7 PSI)(700 - 70 F) = 1.73e^5 PSI$

$$\sigma_\theta = \frac{Pr}{t} = \frac{12.17 PSI(69.5 In)}{0.5 In} = 1690 PSI$$

$$F_{Axial} = PA = P_{avg}(2\pi r) = \frac{12.17 + 47.72 PSI}{2}(2\pi(69.5^2 - 41.75^2 In)) = 2.88e^5 LBF$$

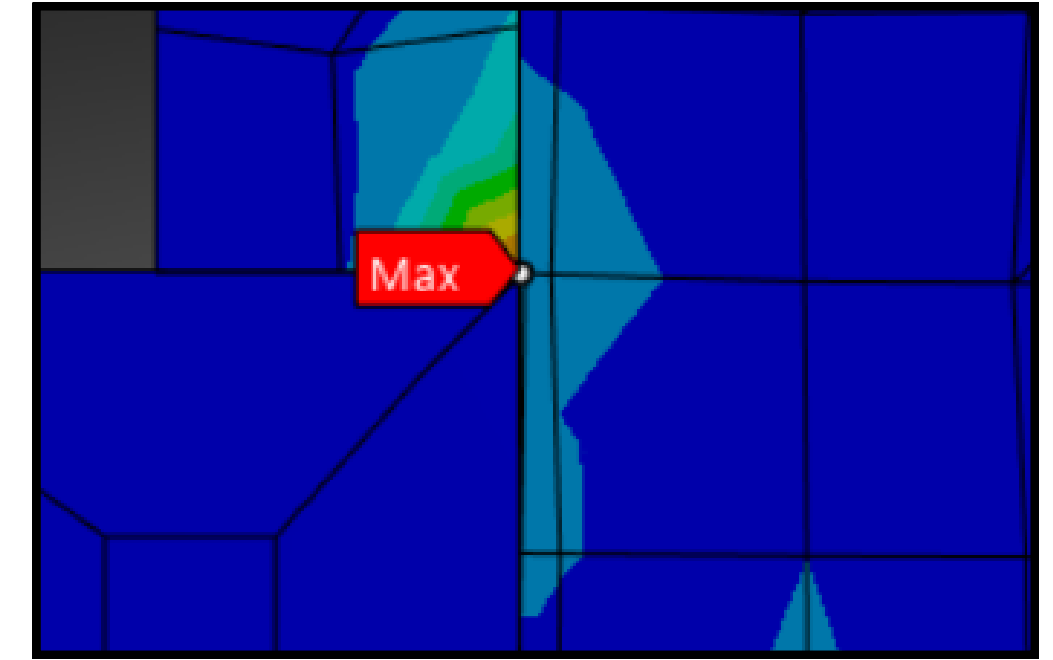
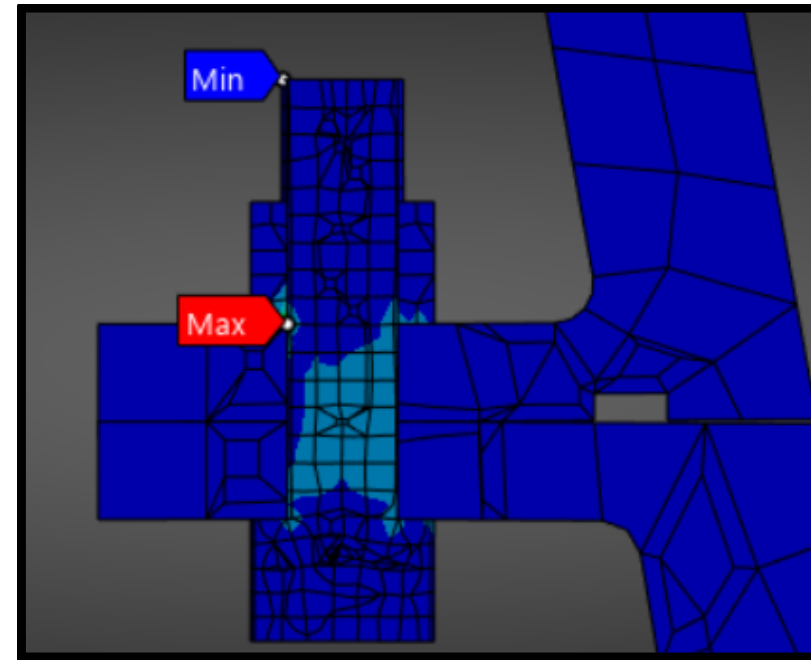
F_{axial} is setup as only half a bolt in this model out of 200 total bolts.

$$F_{axial, 1} = \frac{2.88e^5 LBF}{400} = 720 lbf$$

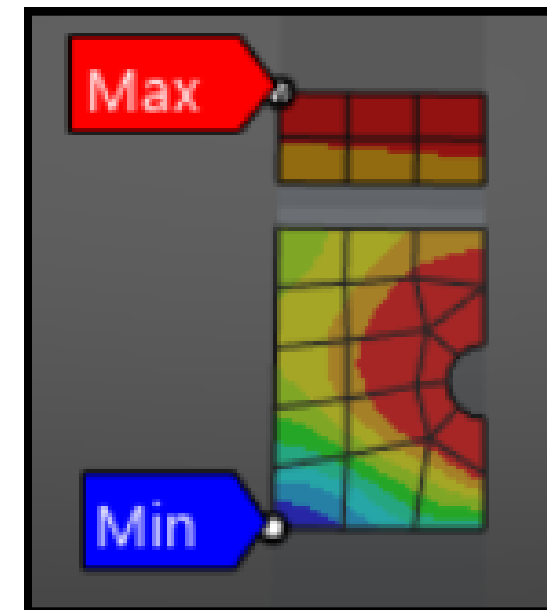
$$\Delta L = \alpha \Delta T L = (9.5e^{-6} 1/F)(700 - 70 F)(160 In) = 1.01 In \text{ or } 0.063 \text{ per unit length}$$

$$\sigma_L = \frac{Pr}{2t \cos(\theta)} = \frac{12.17 PSI(69.5 In)}{2(0.5 \cos(10) In)} = 858.9 LBF \text{ where theta is defined by angle of frustum}$$

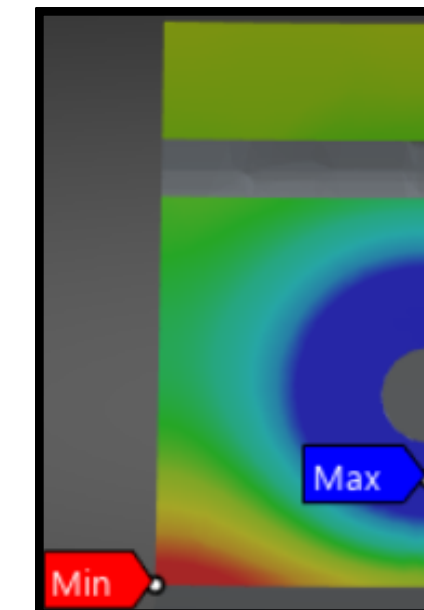
Using deformation, force reaction, stress and strain, Ansys model is verified to be accurate.



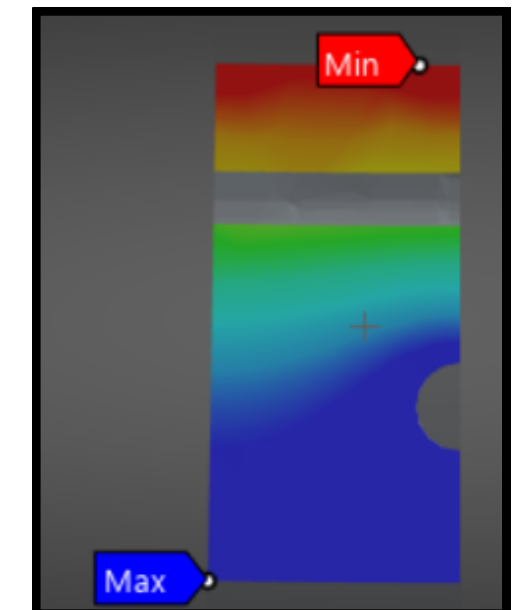
Local mesh convergence study confirmed a stress singularity at a sharp internal corner. Adding a 0.1-inch fillet in CAD successfully resolved the stress.



T=1 Low density



T=1 High density



T=3 High density

Flange separation over time. Results validate bolt pretension causing compression at surface, while comparing mesh density shows low element density causes shear locking.

Brycen Schoenberg